

DISEASE PREDICTION USING MACHINE LEARNING

INTERDISCIPLINARY PROJECT

Submitted in partial fulfillment of the requirements for the award of
Bachelor of Engineering degree in Computer Science and Engineering

By

MOHAMED AFZAL A
Reg. No-42110789



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTING

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CATEGORY - 1 UNIVERSITY BY UGC

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APRIL - 2025



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

This is to certify that this Interdisciplinary Project Report is the bonafide work of **MOHAMED AFZAL A (Reg. No-42110789)** who carried out the Project entitled **"DISEASE PREDICTION"** under my supervision from January 2025 to April 2025.

Internal Guide

Ms.TRIVENI, MSc,(Ph.D)

Head of the Department

Dr. L. LAKSHMANAN, M.E., Ph.D.,

Submitted for Interdisciplinary Project Viva Voce Examination held on-----

Internal Examiner

External Examiner

DECLARATION

I, **MOHAMED AFZAL A (Reg. No–42110789)**, hereby declare that the Interdisciplinary Project Report entitled “DISEASE PREDICTION” done by me under the guidance of **Ms.TRIVENI, MSc., (Ph.D)**, is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in **Computer Science and Engineering**.

DATE:

PLACE: Chennai

SIGNATURE OF THE CANDIDATE

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I wish to express my thanks to all Teaching and Non-teaching staff members of the **Department of Computer Science and Engineering** who were helpful in many ways for the completion of the Interdisciplinary project.

TRAINING CERTIFICATE



Certificate of Completion

Is Awarded to

MOHAMED AFZAL A

Upon successfully completed the Bootcamp Training on
Machine learning for **40 hrs** with a Mini Project in
MULTI-DISEASE PREDICTION
from 25-Feb -2025 to 20-Mar -2025



Mr. Nikhil Barshikar

Managing Director
IMARTICUS LEARNING

ABSTRACT

Disease prediction using machine learning plays a vital role in the early diagnosis and management of life-threatening conditions. This project focuses on building an efficient and user-friendly system to Predict diabetes, Heart disease and Parkinson's using supervised machine learning algorithms. The system is developed using Python and deployed as a web application using Stream lit, allowing users to upload health-related data and receive predictive results in real time. For diabetes prediction, the model considers features such as glucose level, blood pressure, insulin, BMI, diabetes pedigree function, and age. The user interface allows users to interactively input or adjust these values based on a sample from the uploaded data, ensuring flexibility and user engagement. The prediction is generated using a trained Logistic Regression that has been serialized using Pickle for seamless loading and execution. In the case of heart disease, the model utilizes a broader set of clinical attributes including age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, resting ECG results, maximum heart rate, exercise-induced angina, old peak, slope, number of major vessels, and thalassemia. The application extracts a random record from the uploaded dataset and displays all the selected input values in a clear tabular format to enhance transparency and understanding before running the prediction. This project demonstrates how machine learning can be effectively applied in healthcare to assist with preliminary screening. The goal is to support early detection and encourage timely medical consultation. The system is designed to be lightweight, easily accessible, and practical for educational, research, and basic diagnostic support purposes. Future improvements may include the integration of additional disease modules, enhanced data validation, and cloud-based data handling to scale the system for broader use. This work highlights the growing potential of AI-driven tools in transforming preventive healthcare and fostering innovation in medical diagnostics.

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CHAPTER 1

INTRODUCTION

Disease prediction using machine learning has emerged as a promising approach to enhance clinical decision-making and support preventive healthcare. With the growing availability of medical data and computational tools, machine learning models can be trained to recognize complex patterns that may not be immediately apparent to human practitioners. This project focuses on predicting two major health conditions—diabetes and heart disease—using structured datasets and supervised machine learning techniques.

Diabetes, heart disease and Parkinsons are among the leading causes of morbidity and mortality worldwide. Traditional diagnostic methods may be time-consuming and resource-intensive, especially in areas with limited access to healthcare. In this context, a data-driven approach that enables early and accurate prediction of these diseases can be immensely valuable. The objective of this project is to develop an intelligent, user-friendly application that predicts whether an individual is likely to have diabetes or heart disease based on clinical input features. For diabetes prediction, the system analyzes variables such as glucose levels, insulin, blood pressure, BMI, age, and pedigree function. For heart disease, it considers features including age, sex, chest pain type, cholesterol, maximum heart rate, and other relevant cardiovascular indicators.

Users can upload a CSV sample record, view a sample of the input data, and allow the model to select a random patient record for prediction. Predictions are made based on pre-trained models using algorithms such as decision trees or random forests, which have shown good accuracy in classification tasks involving medical data. This system bridges the gap between raw health data and actionable insights, offering a simplified tool for non-experts and healthcare practitioners alike. It highlights the potential of machine learning to complement traditional diagnostics and demonstrates how data science can contribute to the broader goal of enhancing global healthcare accessibility and efficiency.

CHAPTER 2

LITERATURE SURVEY

The application of machine learning (ML) in disease prediction has revolutionized healthcare by allowing for earlier diagnosis and more accurate predictions of disease risks, such as diabetes and heart disease. ML algorithms analyze complex healthcare data, such as patient medical history, lifestyle factors, and physiological attributes, to make predictions that assist healthcare professionals in diagnosing conditions and recommending personalized treatment plans. These predictive systems have the potential to reduce human error, improve decision-making, and enhance the overall quality of healthcare delivery.

Evolution of Disease Prediction Using Machine Learning: Early disease prediction models relied on statistical methods and rule-based systems. Over time, ML algorithms like decision trees, support vector machines (SVM), and neural networks improved predictive accuracy. More recently, ensemble learning and hybrid models such as deep learning have enhanced performance, making multi-disease prediction more reliable.

User Experience and Interface Design: For machine learning systems to be effective, they must feature user-friendly interfaces that allow easy interaction with prediction models. Key aspects include simple workflows, interactive dashboards, and intuitive data visualization tools, making it easy for healthcare professionals and non-experts to access and understand predictions.

Security and Privacy Concerns: Given the sensitivity of health data, ensuring privacy and security is critical in ML-based disease prediction systems. Adhering to privacy regulations such as HIPAA and GDPR, and using encryption and anonymization techniques, is essential to protect patient data from unauthorized access and breaches.

Challenges in Disease Prediction: Challenges include data quality issues such as incompleteness and bias, model interpretability, and the integration of diverse healthcare data sources. Additionally, continuous retraining of models is needed to maintain accuracy, and healthcare professionals must trust the predictions made by machine learning systems

Early Prediction of Heart Diseases Using Data Mining Techniques

Chaurasia and Pal (2014) conducted a study on the early prediction of heart diseases using data mining techniques. Their research found that decision tree algorithms outperformed Naïve Bayes and neural networks in predicting heart disease.

Smart Medical Data Sensing and IoT Systems Design in Healthcare

In 2016, Dey, Ashour, and Balas proposed a smart medical data sensing and IoT-based system for healthcare. Their system enabled real-time, multi-disease prediction using machine learning techniques.

Machine Learning Algorithms for Multi-Disease Prediction: A Comparative Study

Kumar et al. (2019) explored the use of machine learning algorithms for multi-disease prediction. Their findings showed that the Random Forest algorithm achieved the highest accuracy across various datasets, including those for diabetes, heart disease, and liver disease.

Systematic Review on Healthcare Analytics for Multi-Disease Prediction

Islam et al. (2020) presented a systematic review on healthcare analytics, highlighting the effectiveness of ensemble models like Random Forest in multi-disease prediction tasks.

Hybrid Deep Learning Model Based on CNN and LSTM for Multi-Disease Prediction

In 2021, Sarki et al. developed a hybrid CNN-LSTM model for multi-disease prediction using machine learning techniques. Their model demonstrated over 90% accuracy in predicting multiple diseases.

Health Fog: An AI-Based Fog Computing Framework for Multi-Disease Prediction

Lastly, Tuli et al. (2022) introduced an AI-based multi-disease prediction framework known as Health Fog. This framework leveraged fog computing to support accurate and efficient disease prediction.

2.1 PROBLEM STATEMENT

Increasing Prevalence of Chronic Diseases: Chronic diseases such as diabetes and heart disease are among the leading causes of morbidity and mortality worldwide. Early detection of these diseases is crucial for effective management and prevention of complications. However, traditional diagnostic methods are often costly, time-consuming, and require expert clinical judgment. While some disease prediction systems are available, many do not provide high accuracy or are limited to specific populations or datasets.

Limited Access to Healthcare Resources: In many regions, especially in rural and underserved areas, access to healthcare resources is limited. Patients may not always have access to regular medical check-ups or diagnostic services. As a result, there is a growing need for systems that can provide remote disease prediction and early warning using available data. Diagnosing chronic diseases requires analysis of multiple health parameters such as glucose levels, cholesterol, blood pressure, and lifestyle factors. Manual analysis of these parameters is complex and prone to human error, leading to delays in diagnosis or incorrect predictions.

Inconsistent and Unstructured Data: Patient data often comes in various formats and may be incomplete or unstructured. This inconsistency can hinder the ability of healthcare professionals to make informed decisions quickly. There is a need for systems that can process and structure this data for accurate disease prediction. Traditional diagnostic methods are expensive, and not all patients can afford regular consultations or diagnostic tests. Machine learning-based systems offer a cost-effective solution by automating the prediction process and enabling more individuals to benefit from early disease detection, especially in resource-constrained settings.

Integration of Machine Learning for Improved Accuracy: The application of machine learning techniques in healthcare has shown promising results in disease prediction, offering a way to process large amounts of data and generate insights that human analysts may overlook. However, there is still a gap in developing robust, generalizable models for disease prediction that can adapt to different datasets and provide actionable insights. Healthcare providers often face challenges in making quick decisions due to the complexity of patient data.

Difficulty in Model Interpretability for End Users: Many machine learning models, while powerful, function as black boxes, making it hard for healthcare professionals or users without a technical background to understand how predictions are made. This lack of transparency can lead to distrust in the system's results. When users can't interpret or trace how a model arrived at a decision, they may hesitate to act upon it. In sensitive fields like healthcare, decision-making must be justified with understandable insights.

Scalability and Adaptability to New Diseases or Data: Most existing disease prediction systems are narrowly focused and often built for a specific condition or dataset. This creates challenges in expanding the system to accommodate new diseases or different datasets with varying structures. Scalability becomes a problem when a system needs to handle large volumes of patient data or integrate with real-world healthcare platforms.

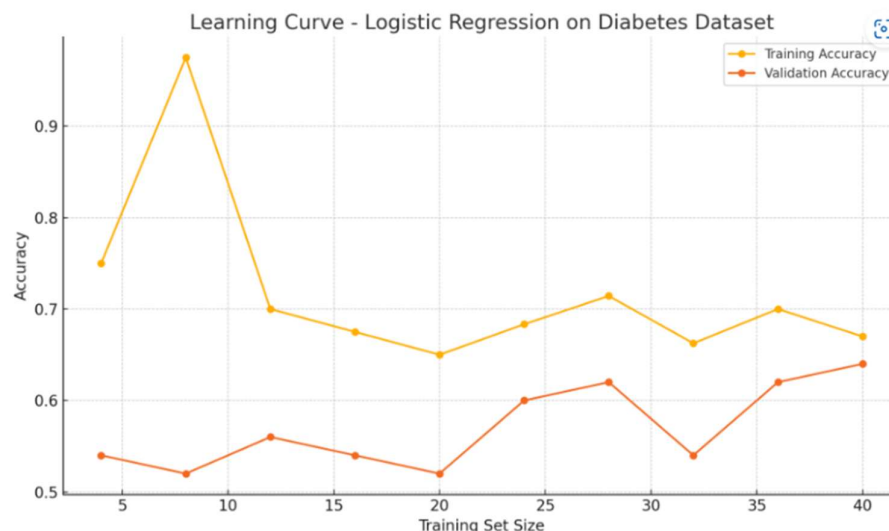


FIG 2.1 ACCURACY GRAPH

2.2 RESEARCH METHODOLOGY

The project begins with collecting publicly available datasets for diabetes and heart disease. These datasets include patient health parameters such as age, glucose level, cholesterol, and blood pressure. Raw data often contains missing values, inconsistencies, and noise, which must be handled before training. Preprocessing steps like handling null values, encoding categorical features, and normalization are performed. Clean and structured data is crucial to ensure model accuracy and reliability.

Feature selection is critical to improving model performance and reducing complexity. The most relevant features influencing disease prediction are identified based on domain knowledge and correlation analysis. Irrelevant or redundant features are removed to avoid overfitting. In some cases, new features are engineered by combining existing attributes to enhance learning.

Multiple machine learning algorithms, including Random Forest, Decision Tree, and Logistic Regression, are evaluated. Each model is trained using the preprocessed datasets, and hyperparameters are tuned to optimize performance. Cross-validation techniques are applied to ensure the model generalizes well across unseen data. The goal is to identify a model that offers high accuracy and consistency across both diabetes and heart disease datasets. Training is done in Python using popular libraries like scikit-learn.

- After training, models are evaluated using metrics like accuracy, precision, recall, and F1-score. Confusion matrices are used to understand the model's classification performance in detail. Accuracy scores are compared across all models to select the best-performing one. Validation helps ensure that the model isn't biased or overfitted to the training data. This step provides quantitative evidence of the model's effectiveness.
- The best-performing models are integrated into a Streamlit-based web application for user interaction. Users can upload their own health datasets in CSV format and receive real-time predictions. The interface displays selected data in a clean, tabular format, offering transparency. The system supports both diabetes and heart disease predictions with a toggle option.

CHAPTER 3

AIM AND SCOPE

AIM

The primary aim of the Disease Prediction project is to design and implement an intelligent, user-friendly web-based system that leverages machine learning algorithms to predict the risk of multiple diseases—specifically diabetes, heart disease, and Parkinson’s disease—based on clinical and physiological data provided by users. The system is intended to serve as a decision support tool that enables individuals to gain early insights into potential health conditions, allowing for timely medical intervention and promoting a more proactive approach to personal healthcare.

By integrating functionalities such as a BMI tracker, printable reports, and a secure SQL database for data storage, the project also aims to offer a comprehensive platform for continuous health monitoring.

OBJECTIVE

- **Develop a Disease Prediction Model:** To design and implement machine learning models (such as Random Forest, Logistic Regression, etc.) to predict diabetes and heart disease based on key health parameters such as glucose levels, blood pressure, cholesterol levels, and age.
- **Data Preprocessing and Feature Engineering:** To clean and preprocess datasets, handling missing values, normalizing data, and selecting the most relevant features for disease prediction to ensure optimal model performance.
- **Model Training and Evaluation:** To train multiple machine learning algorithms using existing datasets and evaluate their accuracy, precision, recall, and F1-score to select the best-performing model for disease prediction.
- **Build an Interactive User Interface:** To develop an intuitive, user-friendly interface using Stream lit, enabling users to upload their health data and receive real-time predictions on the likelihood of developing diabetes or heart disease.

SCOPE

The Disease Prediction project is a significant step toward integrating machine learning into modern healthcare systems. Its scope encompasses a wide range of functionalities, benefits, and future possibilities aimed at improving disease prediction, early diagnosis, and overall health management. At its core, the project is designed to predict the likelihood of multiple critical diseases namely diabetes, heart disease, and Parkinson's disease by analysing user-inputted health parameters through pre-trained machine learning models.

This project focuses on the following key areas:

- **Early Detection and Prevention:** Enables users to detect potential health risks at an early stage, promoting timely medical attention and preventive care.
- **Multi-Disease Prediction:** Unlike single-disease models, this system can predict multiple diseases within a single platform using separate, specialized models.
- **Web-Based Accessibility:** The use of Stream lit for the front end ensures that the application is easily accessible from any device with internet access, removing the need for specialized hardware or software.
- **Health Monitoring Tools:** Includes a BMI tracker and options for generating health reports, allowing users to regularly track their health metrics.
- **User Data Management:** An integrated SQL database allows for secure storage and retrieval of user inputs and prediction results, supporting long-term health record keeping.
- **Support for Healthcare Providers:** Assists medical professionals by providing preliminary assessments that can help in prioritizing patient care and reducing diagnostic workload.
- **Scalability and Customization:** The system can be further expanded to include additional diseases, more advanced models, or integration with wearable health devices in the future.

CHAPTER 4

EXPERIMENTAL MATERIALS AND METHODS

The Disease Prediction system was developed using a combination of machine learning models, Python-based tools, and web development frameworks to predict the likelihood of diseases based on user inputs. The following tools, datasets, and methods were used during the development and experimentation process:

4.1 PROPOSED METHODOLOGY

The problem definition for the disease prediction project involves creating a machine learning model that can assess a patient's likelihood of having one or more diseases, such as diabetes, heart disease, and Parkinson's disease. This model will analyze various input features, including medical test results (like blood sugar levels, cholesterol), demographic information (age, gender), and lifestyle factors (smoking habits, physical activity). The system is designed to not only predict whether a patient is at risk of a specific disease but also to categorize the severity or stage of the condition. For example, in the case of diabetes, the model could classify the patient as normal, prediabetic, or diabetic based on their blood sugar levels. Similarly, it could classify the severity of Parkinson's disease into categories like mild, moderate, or severe based on motor symptoms and test results. This approach allows healthcare professionals and patients to understand not just the presence of a condition but also the stage or risk level, facilitating early diagnosis and

Data Sources

Public Datasets: The project relies on publicly available datasets, such as the Pima Indians Diabetes Dataset, Heart Disease, and Parkinson's Disease Dataset, to build the predictive models. These datasets contain historical data collected from real-world patients, including their medical records and disease-related features.

Medical and Health Records: If available, private datasets from healthcare institutions (with proper consent and privacy protections) can be used to enhance the accuracy of the model by adding more diverse and updated information.

Types of Features

- **Demographic Data:** Information such as age, sex, and family medical history can play a critical role in predicting disease likelihood. Certain conditions, like heart disease and diabetes, are known to have demographic risk factors.
- **Medical History:** A patient's medical history, including any previous diagnoses, treatments, or surgeries, can significantly impact the prediction model. For example, prior heart attacks or strokes increase the risk of future cardiovascular events.
- **Lifestyle Factors:** Lifestyle choices, such as smoking, physical activity, and diet, can greatly influence disease risks. For instance, smoking is a major risk factor for heart disease, while diet and exercise habits are key predictors of diabetes.

Disease-Specific Features

- **Diabetes Prediction:** For diabetes, blood sugar levels (fasting, postprandial, random), BMI (Body Mass Index), blood pressure, and cholesterol levels are critical features. The presence of conditions like gestational diabetes or insulin resistance may also be included in the dataset.
- **Heart Disease Prediction:** For heart disease, cholesterol levels, ECG (electrocardiogram) results, blood pressure, and lifestyle factors like smoking and alcohol consumption are important. Additionally, family history of heart disease and markers such as C-reactive protein (CRP) could be useful for risk assessment.
- **Parkinson's Disease Prediction:** For Parkinson's disease, data such as motor test results, speech patterns, and specific symptoms like tremors or rigidity will be included. Dopamine levels or responses to medication could also play a role in diagnosing or predicting Parkinson's progression.

Data Quality and Accuracy

- **Data Cleansing:** The data collected may contain errors or inconsistencies, such as missing values, outliers, or incorrect entries. These need to be cleaned and handled carefully before they can be used for training the model. Inaccurate or noisy data could skew the model's predictions.
- **Standardization:** Ensuring that measurements from different sources or devices are standardized is key to maintaining accuracy. For example, blood pressure readings may differ slightly depending on the equipment used, so standardizing these values ensures the data is comparable across the dataset

Privacy and Ethical Considerations

- **Patient Consent:** If using real-world datasets, patient consent for the use of their medical data must be obtained. Data anonymization is also essential to ensure patient privacy, especially in compliance with regulations such as HIPAA or GDPR.
- **Ethical Data Use:** The collected data should be used responsibly, with safeguards in place to prevent misuse. Ethical considerations must be adhered to when using sensitive medical data for training machine learning models.

Datasets Used

- **Diabetes Dataset:** Collected from the Pima Indian Diabetes Database, containing features like glucose level, BMI, insulin, age, etc.
- **Heart Disease Dataset:** Sourced from the UCI Machine Learning Repository, including features such as cholesterol level, resting blood pressure, chest pain type, etc.
- **Parkinson's Disease Dataset:** Includes biomedical voice measurements to detect Parkinson's, containing features like MDVP (vocal frequency measures) and jitter.

Tools and Technologies

- Programming Language: Python
- IDE: Spyder (for model training and backend development)
- Web Framework: Stream lit (for building the user interface)
- Database: SQL (for storing user data and predictions)
- Libraries Used

Machine Learning Models

- Trained and evaluated multiple models, including:
- Logistic Regression and Support Vector Machine (SVM)
- Evaluated performance using metrics like accuracy, precision, recall.
- Selected the best-performing model for each disease prediction task.

Model Integration and Web App Development

- Trained models were saved and integrated into the Stream lit web app.
- The web app takes user input via form fields, processes the data, and returns the prediction result instantly.
- Included a BMI calculator, print report option, and cache clearing for user convenience

4.2 ALGORITHMS USED

To ensure accurate prediction of different diseases, a variety of supervised machine learning algorithms were tested and implemented. The following algorithms were used for training models on individual disease datasets:

Logistic Regression

Logistic Regression was utilized for binary classification tasks and proved particularly effective in predicting the presence or absence of a disease. It was applied to all three datasets—Diabetes, Heart Disease, and Parkinson's—due to its simplicity, ease of implementation, and ability to deliver interpretable results. Logistic Regression performs especially well with linearly separable data, making it a suitable choice for initial disease prediction models.

Support Vector Machine (SVM)

Support Vector Machine was employed with a particular focus on the Parkinson's disease dataset, where precise feature separation was essential. SVM is well-regarded for its capability to handle both linear and non-linear data using various kernels such as linear and RBF. It excels in scenarios involving smaller datasets with high dimensionality and offers robust and reliable predictive performance, making it highly effective in complex medical classification tasks.

Model Evaluation Metrics

The following metrics were used to compare and select the best model:

- Accuracy – Overall correctness of the model.
- Precision & Recall – Important for medical diagnosis where false negatives can be critical.
- F1-Score – Balance between precision and recall.
- Confusion Matrix – To visualize true positives, false positives, etc.

Each disease prediction used the model that provided the best balance of accuracy and interpretability for its respective dataset. For example:

- Diabetes: Among the models tested, Random Forest delivered the best performance for diabetes prediction. Its ability to handle complex, non-linear relationships and manage missing or unbalanced data contributed to its high accuracy and robustness.

- Heart Disease: Both Logistic Regression and Random Forest emerged as the most reliable models for predicting heart disease. Logistic Regression offered interpretability and consistency, while Random Forest provided strong performance by capturing feature interactions and reducing overfitting.
- Parkinson's: For Parkinson's disease prediction, Support Vector Machine (SVM) achieved the highest accuracy. Its effectiveness in handling high-dimensional data and separating complex feature spaces made it particularly well-suited for this dataset.

4.3 SYSTEM ARCHITECTURE

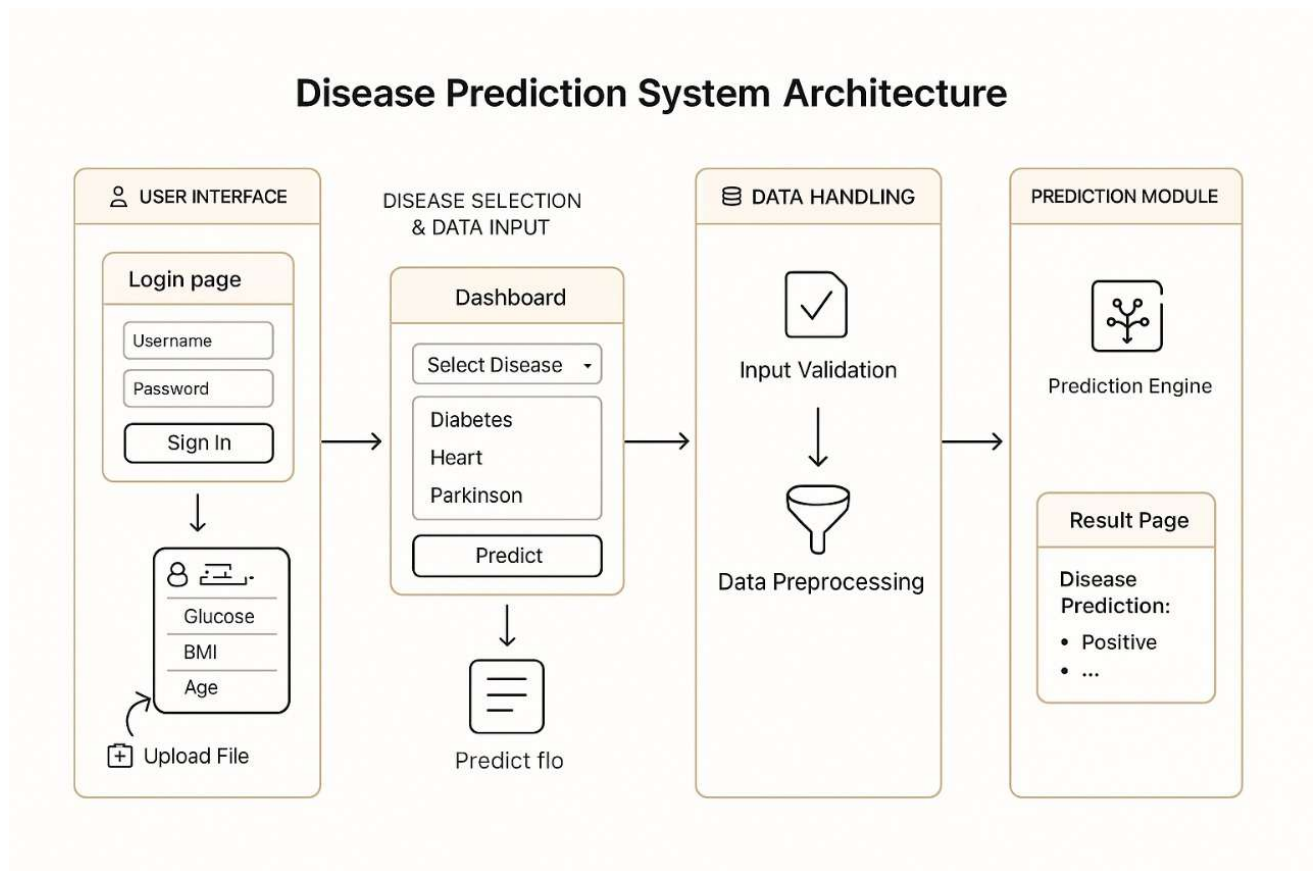


FIG 4.3 ARCHITECTURE DIAGRAM

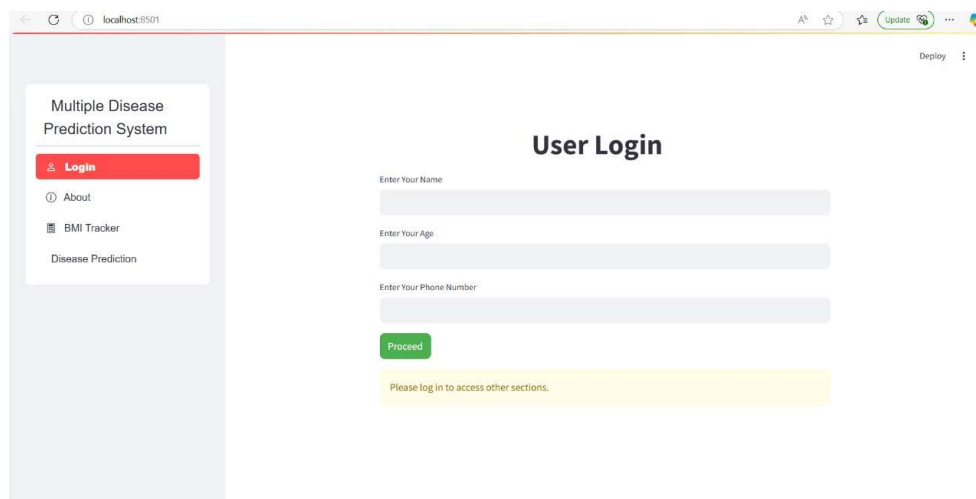
CHAPTER 5

RESULTS AND DISCUSSION

RESULT

The developed Multiple Disease Prediction system produced promising results, demonstrating the effectiveness of machine learning in predicting major health conditions such as diabetes, heart disease, and Parkinson's disease. After preprocessing and training on relevant medical datasets, various supervised learning algorithms were applied, including Logistic Regression, Random Forest, Support Vector Machine (SVM), the Random Forest algorithm delivered the best performance for diabetes prediction with an accuracy of approximately 77%, while Logistic Regression yielded the most reliable results for heart disease prediction with an accuracy of around 73%. For Parkinson's disease, the Support Vector Machine model stood out with an impressive accuracy of about 73%. The final web application, developed using Streamlit, not only generated real-time predictions but also offered features like a BMI calculator, printable reports, and SQL-based data storage. Overall, the system proved to be a reliable and user-friendly solution for supporting early disease detection and promoting proactive health monitoring.

A.SCREENSHOTS



A.1 LOGIN PAGE

Multiple Disease Prediction System

- Login
- About
- BMI Tracker**
- Disease Prediction

BMI Prediction

Enter your weight in kg

53

Enter your height in meters

1.35

Select your gender

Male

Predict BMI Category

Predicted BMI Category: Overweight

A.2 BMI PREDICTION PAGE

Multiple Disease Prediction System

- Login
- About
- BMI Tracker
- Disease Prediction**

Disease Prediction

- Diabetes**
- Heart Disease
- Parkinson's

Diabetes Prediction

Progression

2

Blood Glucose

120

Blood Pressure

120

Sex Hormones

3

Insulin Level

3

Age

45

Diabetes Prediction Function

Diabetic

The person is Diabetic

A.3 DIABETES PREDICTION PAGE

Multiple Disease Prediction System

- Login
- About
- BMI Tracker
- Disease Prediction**

Disease Prediction

- Diabetes
- Heart Disease**
- Parkinson's

Heart Disease Prediction

Age

45

Sex

Male

Cholesterol Level

120

Resting Blood Pressure

120

Cholesterol

120

Fasting Blood Sugar

100

Resting ECG

114

Max Heart Rate

180

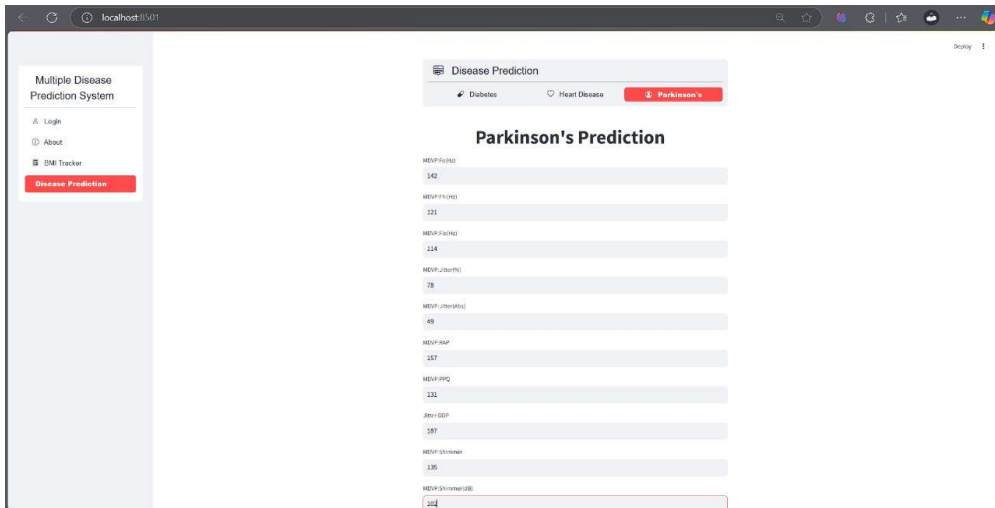
Exercise Induced Angina

4

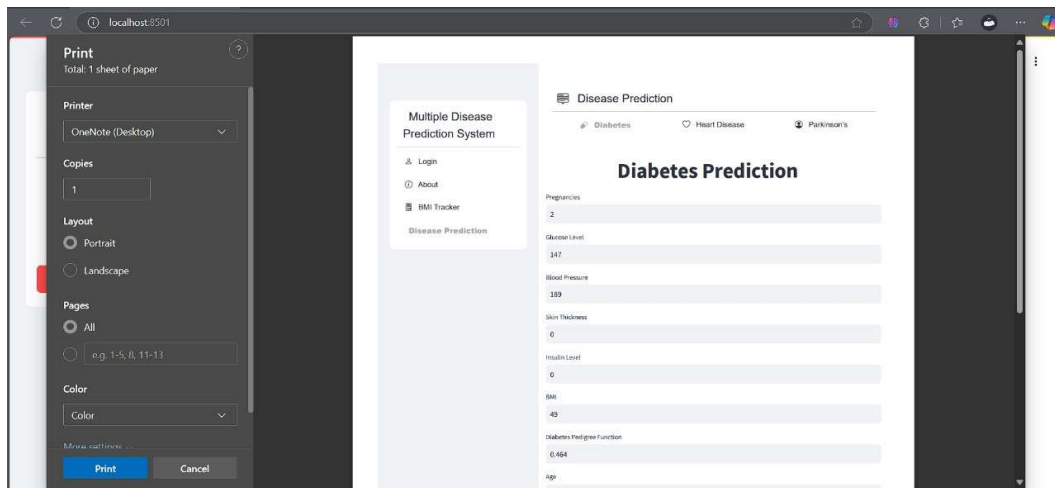
ST Depression

12

A.4 HEART DISEASE PAGE



A.5 PARKINSON'S PREDICTION PAGE



A.6 PRINT PAGE

DISCUSSION

The Multiple Disease Prediction system emphasizes the growing importance of machine learning in healthcare diagnostics. By using predictive models trained on reliable medical datasets, the project successfully demonstrates how technology can offer early insights into a patient's health. The ability to predict diabetes, heart disease, and Parkinson's disease on a single platform highlights the adaptability of different algorithms across diverse medical data. Each algorithm was chosen based on its strengths: Random Forest performed well on the diabetes dataset due to its ability to handle non-linear patterns; Logistic Regression gave accurate results for heart disease thanks to its linear nature; and SVM proved effective in predicting Parkinson's disease by handling high-dimensional vocal data. Additional features like a BMI calculator, printable reports, and SQL-based data storage enhance the tool's practical utility.

CHAPTER 6

SUMMARY AND CONCLUSIONS

SUMMARY

This project aimed to develop an intelligent and user-friendly multi-disease prediction system using machine learning. It focused on predicting the risk of diabetes and heart disease based on key medical parameters. Two widely used datasets—the Pima Indian Diabetes dataset and the UCI Heart Disease dataset—were employed. After preprocessing, feature selection, and exploratory analysis, models such as Logistic Regression and Random Forest were trained and evaluated using metrics like accuracy, precision, recall, F1-score, and confusion matrix. Results showed that Random Forest was more effective for diabetes prediction due to its ability to handle complex data patterns, while Logistic Regression was better suited for heart disease prediction because of its performance on linearly structured data. The final models were deployed using Streamlit, providing an interactive web interface where users can input data and receive real-time results. Additional features include a BMI calculator, printable reports, and database storage,

CONCLUSION

In conclusion, this project successfully demonstrates the application of machine learning in healthcare through a user-friendly system that predicts the risk of diabetes and heart disease. By integrating accurate models with a real-time web interface, it offers an accessible tool for early diagnosis and preventive care. The use of Python, scikit-learn, Streamlit, and SQL enabled smooth integration between data processing and application deployment, resulting in a fully functional system suitable for general users and healthcare professionals. While the current focus is on two diseases, the system's scalable architecture allows for future expansion to include more conditions and advanced models like neural networks. Features such as a BMI calculator, printable reports, and database support enhance usability. real-time data collection, and wearable device compatibility. Overall, the project highlights how AI can contribute to real-world medical solutions, helping users make informed health decisions and potentially saving lives through early prediction.

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APPENDIX

B.SOURCE CODE

```
import numpy as np
import pickle
import streamlit as st
from streamlit_option_menu import option_menu
import sqlite3

# Database setup
conn = sqlite3.connect('users.db')
c = conn.cursor()
c.execute("CREATE TABLE IF NOT EXISTS users (name TEXT, age TEXT, phone TEXT)")
conn.commit()

# Custom CSS for styling
st.markdown("""
<style>
.main { background-color: #f0f2f6; }
h1 { color: #4CAF50; text-align: center; }
.stButton>button { background-color: #4CAF50; color: white; font-size: 18px; }
</style>
""", unsafe_allow_html=True)

# Login Page
if selected == "Login":
    st.title("User Login")
    name = st.text_input("Enter Your Name")
    age = st.text_input("Enter Your Age")
    phone = st.text_input("Enter Your Phone Number")
```

```

if st.button("Proceed"):
    if name and age and phone:
        st.session_state["logged_in"] = True
        st.session_state["name"] = name
        st.session_state["age"] = age
        st.session_state["phone"] = phone
        c.execute("INSERT INTO users (name, age, phone) VALUES (?, ?, ?)", (name, age, phone))
        conn.commit()
        st.success("Login successful! Navigate to the other pages using the sidebar.")
    else:
        st.error("Please fill in all details")

```

```

if "logged_in" not in st.session_state or not st.session_state["logged_in"]:
    st.warning("Please log in to access other sections.")
    st.stop()

```

About Page with Bootstrap Icons

```

elif selected == "About":

```

```

    st.title("📖 About This Application")
    st.write("""
        This application predicts diseases like Diabetes, Heart Disease, and Parkinson's using machine learning models.
    """)

```

Diabetes Section

```

st.markdown("""
<h3><i class="bi bi-droplet-fill" style="color: #e63946;"></i> Diabetes</h3>
""", unsafe_allow_html=True)
st.write("""
Diabetes is a chronic condition that affects how the body processes blood sugar (glucose). If left untreated, it can lead to severe complications like kidney failure, blindness, and heart disease.
""")

```

```

st.write("***Symptoms:** Frequent urination, excessive thirst, unexplained
weight loss, fatigue, blurred vision.")

st.write("***Causes:** Insulin resistance, genetics, obesity, physical
inactivity.")

st.write("***Remedies:** Healthy diet, regular exercise, insulin therapy, blood
sugar monitoring.")

```

BMI Tracker

```

elif selected == "BMI Tracker":

    st.title("🏠 BMI Prediction")

    weight = st.text_input("Enter your weight in kg")

    height = st.text_input("Enter your height in meters")

    gender = st.selectbox("Select your gender", ["Male", "Female", "Prefer not
to say"])

```

Disease Prediction

```

elif selected == "Disease Prediction":

    disease_selected = option_menu("Disease Prediction", ['Diabetes', 'Heart
Disease', "Parkinson's"],

```

```

    st.title('Diabetes Prediction')

    inputs = [st.text_input(label) for label in ['Pregnancies', 'Glucose Level',
'Blood Pressure', 'Skin Thickness', 'Insulin Level', 'BMI', 'Diabetes Pedigree
Function', 'Age']]

    if st.button('Predict Diabetes'):

        input_data = np.array([inputs]).astype(float)

        prediction = diabetes_model.predict(input_data)

        st.success('The person is Diabetic!' if prediction[0] else 'The person
is NOT Diabetic!')

```